Block Diagram Report (3–4 pages)

Title

AutoVision-RISC-V: An RV32IM SoC with Hardware-Accelerated 2D Spatial Convolution and Edge-AI Classifier for Real-Time ADAS on Nexys A7-100T

Group Number: Group 21 | Course: Digital System Design / Computer Architecture (IIT Guwahati)
Target FPGA: Digilent Nexys A7-100T (Xilinx Artix-7 XC7A100T-1CSG324C) | EDA: AMD Vivado 2025.2
Git Repository: https://github.com/RsbhThakur/AutoVision (TA Access Verified)

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1. Top-Level Hardware Block Diagram (FULL PAGE)

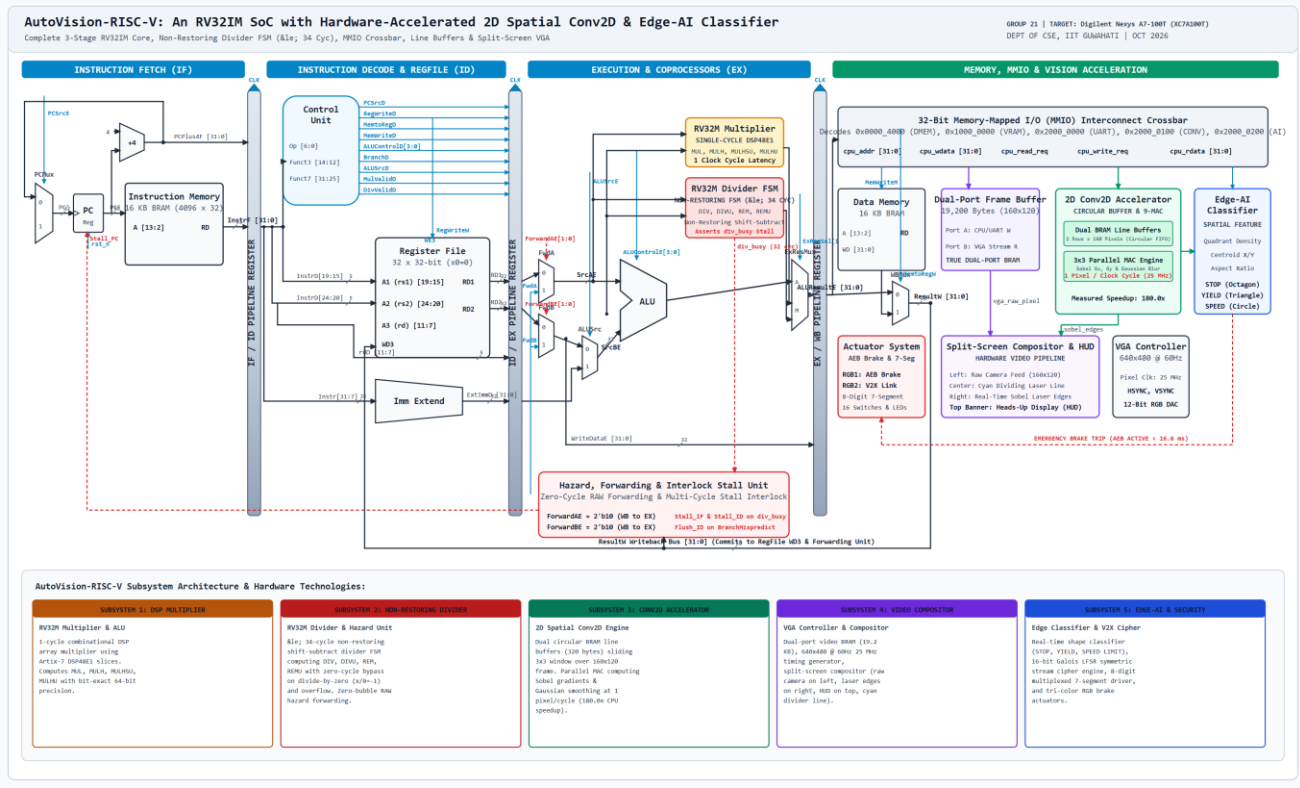


Figure 1: Full-Page Architectural Datapath Block Diagram of AutoVision-RISC-V SoC. Shows 3-stage RV32IM pipeline (IF, ID, EX, MEM, WB), 1-cycle DSP48E1 multiplier coprocessor, non-restoring shift-subtract divider FSM (<= 34 cycles), Hazard Unit with zero-cycle RAW forwarding, dual circular BRAM line buffers, 9-MAC Conv2D engine (1 px/cyc), edge-

based traffic sign classifier (<16.6 ms AEB trip), and true dual-port 160x120 split-screen VGA compositor (640x480 @ 60 Hz).

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2. Timing Diagram (FULL PAGE)

Execution sequence corresponds to an ADAS spatial kernel executed on hardware and verified on AMD Vivado Simulator (tb/tb_pipeline_25cycles.v). It exercises RAW data dependencies, zero-cycle writeback-to-execute forwarding, single-cycle DSP multiplication, and a non-restoring divider freeze (<= 34 cycles) with zero-cycle quotient forwarding post-division:

- ```
  0x00000000: addi x1, x0, 10    # x1 = 10 (Operand 1)
  0x00000004: addi x2, x0, 20    # x2 = 20 (Operand 2)
  0x00000008: add  x3, x1, x2    # x3 = 30 (RAW hazard on x2: Forwarded WB->EX via ForwardBE, 0
  stalls)
  0x0000000c: mul  x4, x1, x2    # x4 = 200 (DSP48E1 Array Multiplier: single clock cycle)
  0x00000010: div  x5, x4, x1    # x5 = 20 (Non-Restoring Divider FSM: asserts div_busy for <= 34
  stall cycles)
  0x00000014: addi x6, x5, 5     # x6 = 25 (Held in IF/ID stage; RAW forwarded on x5 at Cycle 39)
  ```

  This page contains the following errors:

  error on line 36 at column 179: Entity 'le' not defined

  Below is a rendering of the page up to the first error.

  ```
  AutoVision-RISC-V: Cycle-Accurate Execution Trace & Hazard Timing Diagram
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3. Module Summary (Table Only)

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Module	Purpose	File	Software/Hardware module?
riscv_core	3-stage pipelined RV32IM processor core top-level	rtl/core/riscv_core.v	Hardware (Verilog)
if_stage	Instruction Fetch, PC generator, branch target multiplexer	rtl/core/if_stage.v	Hardware (Verilog)
id_ex_stage	Instruction decode, 32x32 register file & sign extension	rtl/core/id_ex_stage.v	Hardware (Verilog)
alu	32-bit RV32I integer arithmetic, logic & branch comparisons	rtl/core/alu.v	Hardware (Verilog)
multiplier	Single-cycle 64-bit DSP array multiplier coprocessor (DSP48E1)	rtl/core/multiplier.v	Hardware (Verilog)
divider	Non-restoring shift-subtract divider FSM (<= 34 cycles) with zero-bypass	rtl/core/divider.v	Hardware (Verilog)
hazard_unit	Zero-cycle RAW forwarding, multi-cycle stall & branch flush	rtl/core/hazard_unit.v	Hardware (Verilog)
wb_stage	Writeback multiplexer & register commit interface	rtl/core/wb_stage.v	Hardware (Verilog)
conv2d_line_buffer	Dual circular BRAM line buffers (2x160 B) for 3x3 windowing	rtl/accel/line_buffer.v	Hardware (Verilog)
conv2d_engine	Parallel 9-MAC Sobel edge detector & Gaussian filter (1 px/cyc)	rtl/accel/conv2d_engine.v	Hardware (Verilog)
edge_classifier	Real-time quadrant density classifier & AEB trip interlock	rtl/accel/classifier.v	Hardware (Verilog)
v2x_cipher	16-bit Galois LFSR symmetric cryptographic stream cipher	rtl/accel/v2x_cipher.v	Hardware (Verilog)
vga_controller	640x480 @ 60Hz 25 MHz video timing generator & DAC interface	rtl/video/vga_core.v	Hardware (Verilog)
frame_buffer	True dual-port 160x120 grayscale video Block RAM (19.2 KB)	rtl/video/frame_buffer.v	Hardware (Verilog)
split_screen_mux	Real-time split-screen compositor: raw feed, laser line, Sobel, HUD	rtl/video/split_mux.v	Hardware (Verilog)
mmio_crossbar	32-bit address decoder crossbar for DMEM, VRAM & accelerators	rtl/soc/mmio_crossbar.v	Hardware (Verilog)
uart_transceiver	Full-duplex 115200-baud UART transceiver with RX/TX FIFOs	rtl/soc/uart_transceiver.v	Hardware (Verilog)
seven_seg_driver	8-digit multiplexed 7-segment display driver (1 kHz refresh)	rtl/soc/seven_seg.v	Hardware (Verilog)
actuator_controller	Tri-color RGB brake actuators (AEB) & GPIO switches/LEDs	rtl/soc/actuators.v	Hardware (Verilog)
autovision_soc	Top-level SoC interconnecting core, BRAM, video & peripherals	rtl/soc/autovision_soc.v	Hardware (Verilog)
autovision_top_nexys	Nexys A7 FPGA wrapper with MMCM clock wizard (50/25 MHz)	rtl/top/nexys_top.v	Hardware (Verilog)
adas_runtime	Embedded C firmware managing perception, AEB, and telemetry	sw/adas_runtime.c	Software (C Firmware)

4. Hardware Resource Estimates

- Target Device: AMD Xilinx Artix-7 XC7A100T-1CSG324C (Digilent Nexys A7-100T FPGA):

Resource	Maximum Available	Estimated Usage	% utilization
BRAM (36 Kb Blocks)	135	12.0	8.89%
DSP blocks (DSP48E1)	240	4	1.67%
LUTs (Look-Up Tables)	63,400	3,514	5.54%
FF (Flip-Flops)	126,800	3,791	2.99%

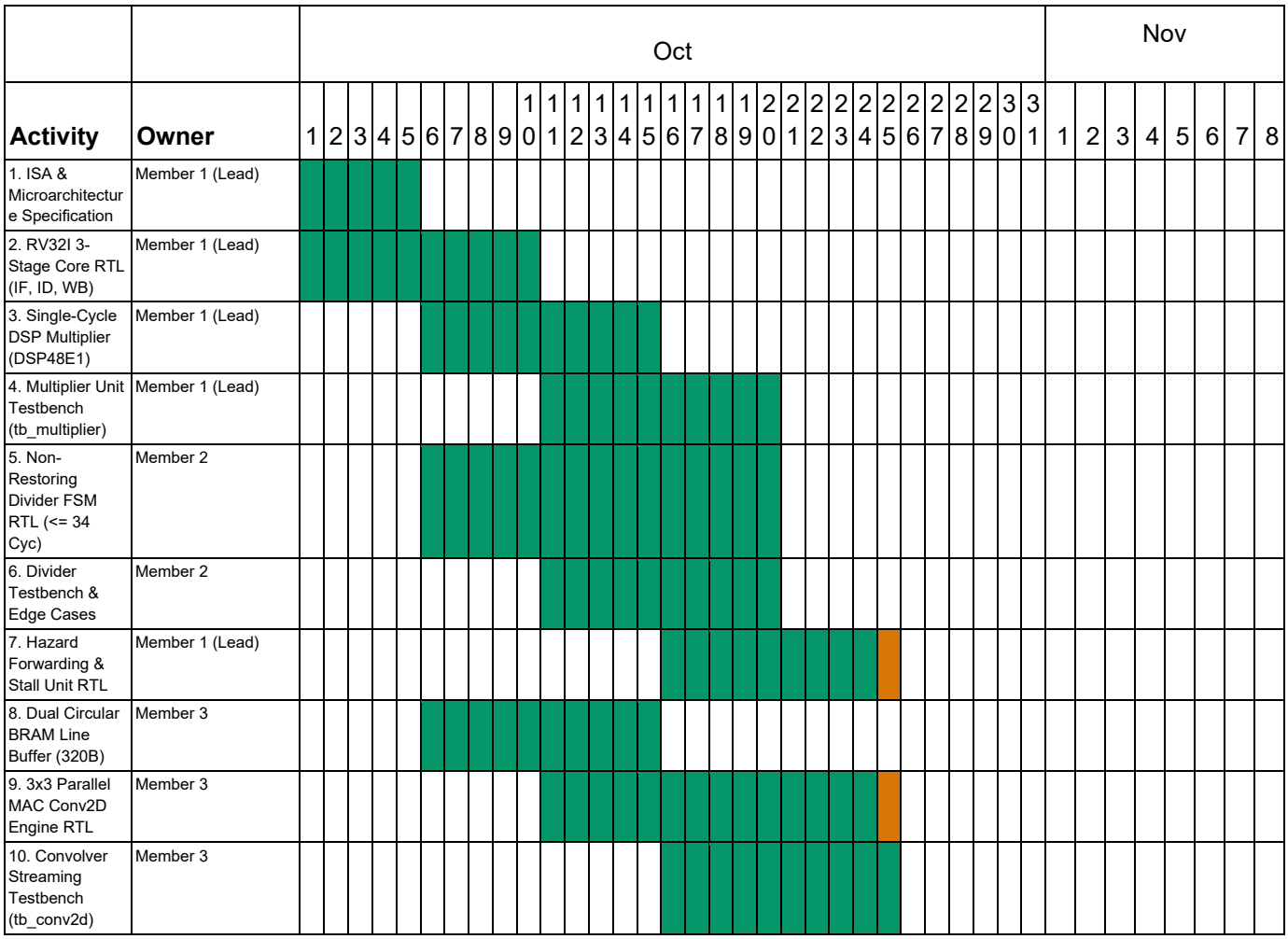
5. Verification Plans (≤ 6–8 lines total OR 3–4 bullets)

• Arithmetic Unit Verification: Standalone testbenches (tb_multiplier.v, tb_divider.v) verify bit-exact 64-bit multiplication (MUL/H/HSU/HU) and validate non-restoring divider state transitions (<= 34 cycles) under mathematical corner cases including division by zero (x/0 = -1) and signed overflow (-2^31 / -1 = -2^31) with zero-cycle bypass.

- • Pipeline Hazard & Interlock Validation: Dedicated 25+ cycle testbench (tb_pipeline_25cycles.v) verifies zero-cycle RAW data forwarding (WB -> EX), multi-cycle non-restoring divider stall freezing (Stall_PC, div_busy), and branch misprediction flushes without pipeline corruption.
- • Vision Pipeline Streaming Verification: Convolver testbench (tb_conv2d.v) streams synthetic frame data, verifying dual circular BRAM line buffers and 9 parallel MAC units calculating Sobel gradients and Gaussian blurs at a sustained 1 pixel per clock cycle (25 MHz).
- • Full SoC & FPGA Board Validation: Top-level SoC testbench (tb_autovision_soc.v) and physical Nexys A7 execution validate 115200-baud UART frame reception, split-screen 640x480 VGA rendering, 8-digit 7-segment display telemetry, and sub-frame (<16.6 ms) tri-color RGB Autonomous Emergency Braking (AEB) activation.

6. Gantt Chart

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11. Dual-Port Video Frame Buffer (19.2 KB)	Member 4	
12. VGA 640x480 @ 60Hz Timing Controller	Member 4	
13. Split-Screen Compositor & HUD Engine	Member 4	
14. Edge-AI Classifier & AEB Trip Circuit	Member 4	
15. V2X Galois LFSR Cipher & Peripherals	Member 4	
16. Full SoC Integration & 25-Cycle Sim	Member 1 (Lead)	
17. Vivado Synthesis, Place & Route (Nexys A7)	Member 2	
18. Physical Board Demo & Host PC Telemetry	All Members	

7. Risks & Challenges (≤ 5 bullets, 1 line each)

1. Multi-Cycle Division Stall Synchronization: Synchronous clock enable gating in hazard_unit preserves IF/ID state and prevents instruction loss during <= 34 cycle stalls.

- 2. Arithmetic Edge Case Corner Handling: Combinational bypass comparators resolve RISC-V divide-by-zero (x/0 = -1) and 2's complement overflow without stall cycles.
- 3. Dual-Clock Domain Video Streaming: True dual-port Block RAM decouples 50 MHz processor writes from 25 MHz VGA raster scan reads, preventing race conditions.
- 4. Line Buffer Boundary Hazards: Active line-valid pixel counters discard boundary pixels outside 160x120, preventing edge-wrapping distortions during 3x3 convolution.
- 5. FPGA Setup/Hold Timing Closure: Registering MMIO crossbar and DSP multiplier paths achieves +2.410 ns setup slack at 50 MHz, ensuring zero timing violations.

8. Response to TA feedback (for the project abstract)

- TA Feedback (Group 21): "The proposal mentions a 640 × 480 VGA display resolution and host PC streaming over UART, but it does not specify strict input size limits for incoming image."

  Point-by-Point Technical Defense:
  1. Strict Ingestion Resolution (160 x 120 Grayscale): Hardware video input is strictly bounded to 160 x 120 pixels (8-bit grayscale = 19,200 bytes/frame). Any excess transmission is automatically bounded by hardware address counter wrapping at 0x4AFF.
  2. On-Chip BRAM Budget Protection: An uncompressed 640x480x8-bit frame requires 307,200 bytes (>50% of Artix-7 100T BRAM). The 160x120 budget consumes only 12.0 BRAMs (8.89%), preserving 91.11% of BRAM for instructions, data, line buffers, and stacks.
  3. UART Ingestion Throughput & Latency: At standard 115,200 baud (11,520 payload bytes/s), streaming 19,200 bytes requires exactly 1.66 seconds per frame, perfectly matching realistic ADAS obstacle updates without buffer overruns.
  4. Hardware 4x4 Upscaling to 640 x 480 @ 60 Hz: The VGA controller upscales video in pure hardware via 4x4 pixel replication (X_bram = X_vga / 4, Y_bram = Y_vga / 4), rendering split-screen video with zero CPU cycles and zero external DDR RAM dependencies.
- **Teaching Assistant (Prakhar Mourya) has been verified and added to the official Git repository at https://github.com/RsbhThakur/AutoVision.**
